

Hyperbolic Latent Geometry for Tree-Structured Prototype Networks: A Local-vs-Global Trade-off

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Abstract

We ask how the choice of latent manifold for class prototypes (Euclidean $\mathbb{R}^d$ vs. the Poincaré ball $\mathbb{B}_c^d$) affects fitting a tree-structured regularizer on prototype layouts in a hierarchical-classification model, since hyperbolic space embeds trees with provably lower distortion than $\mathbb{R}^d$ at matched dimension. Across 150 seed-replicated regularized maximum-likelihood fits on WikiArt (27 styles, 81,446 paintings, frozen CLIP ViT-B/16 features) spanning embedding dimension, curvature, and regularizer strength, Poincaré prototypes preserve the topology of the nearest-neighbor graph in latent space substantially better than matched Euclidean prototypes (sibling recall@5 +8.7 pp, cousin recall +15.2 pp; paired- t $p < 10^{-4}$), and the gap holds across hand-built, CLIP-derived, and DINOv2-derived reference trees. Calibrated against logistic-regression and k -NN baselines on raw encoder features, only the hyperbolic fit improves on the encoder for local retrieval; the Euclidean fit ties logistic regression on classification but adds no detectable structural value over it. Global tree-fidelity comparisons are unstable across reference trees and we do not claim a winner there. The results separate two natural latent geometries for a class-structured regularizer on a real hierarchical-classification problem.

1. Introduction

Many real-world classification problems carry an externally specified hierarchy over class labels: artistic-style movements branch into one another (Tan et al., 2019), biological taxa nest into clades, and ontologies of products, diseases, and documents are organised as trees. A natural way to fold that hierarchy into a probabilistic classification model

is to add a tree-structured regularizer to the layout of class-conditional parameters. We study the simplest version of this: a prototype classifier whose class-conditional likelihood is a softmax over distance-to-prototype, with a regularizer on the matrix of pairwise prototype distances pulling it toward the tree-distance matrix. The fitted classifier is then a balance between a class-likelihood term (paintings of style k should be close to prototype $\mathbf{p}_k$) and a tree-shape term (prototypes themselves should respect the hierarchy).

The choice of latent manifold for the prototypes is the question this paper investigates. Forcing a tree’s exponentially-growing leaf count into Euclidean $\mathbb{R}^d$, whose ball volume grows only polynomially in radius, induces unavoidable distortion of pairwise distances; this distortion has been characterised analytically and shown to drop sharply on the Poincaré ball $\mathbb{B}_c^d$, whose own exponential volume growth matches a tree’s (Nickel & Kiela, 2017; Sala et al., 2018). In our model, the tree-structured regularizer is a function of pairwise prototype distances and should therefore be cheaper to satisfy on the hyperbolic manifold than on the Euclidean one without distorting the class-conditional likelihood. Whether that geometric advantage realises in practice on a real hierarchical-classification problem is what we test.

We instantiate the question on WikiArt-Refined (Tan et al., 2019), a 27-style 81,446-painting corpus where the class hierarchy (Renaissance branches into Baroque, Baroque into Rococo, Rococo into Romanticism, on into Impressionism and Cubism) is part of the domain and was hand-built from standard art-history references. Frozen CLIP ViT-B/16 features (Radford et al., 2021) feed an MLP head and a prototype classifier; the manifold of the classifier output is the only design choice that varies between Euclidean and hyperbolic runs. The same tree-structured regularizer (a normalised squared-error pull on the prototype distance matrix) applies in both geometries.

We sweep regularized maximum-likelihood fits across 150 seed-replicated configurations spanning embedding dimension $d \in \{2, 4, 8, 16, 32, 64\}$, curvature $c \in \{0.1, 0.3, 1, 3\}$, regularizer strength $\lambda \in \{0, 0.1, 0.3, 1, 3\}$, and three reference-tree variants (default lineage, chronological era grouping, flat null). Three empirical findings emerge.

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Local hierarchy. Hyperbolic prototypes recover local tree structure substantially better than matched Euclidean ones. Sibling recall@5 at $d=8$ is 0.195 versus 0.149 on the default tree; aggregated over 18 paired seed configurations the mean gap is +8.7 pp on sibling recall and +15.2 pp on cousin recall, with sign agreement 0.94 and paired- t $p < 10^{-4}$. This effect is preserved when sibling sets are redefined from data-driven empirical trees built either from CLIP or DINOv2 (Oquab et al., 2024) features, and across the regularizer-strength sweep.

Calibration. A logistic-regression baseline on raw CLIP features achieves 64.1% top-1, statistically tied with the Euclidean fit’s 64.3%. A k -NN baseline on the same features achieves sibling recall@5 of 0.160, statistically tied with the Euclidean fit’s 0.149 but 3.5 pp below the hyperbolic fit’s 0.195. Read against these baselines, the hyperbolic fit is the only one of our trained models that improves the encoder on local hierarchy; the Euclidean fit matches the encoder on classification but adds no detectable structural value over it.

Global tree fidelity. Mean tree distortion against the default tree favours Euclidean significantly, but prototype-tree Spearman is not significantly different from zero across the full sweep ($p = 0.68$), and the small effect that does appear flips sign between empirical reference trees built from CLIP and from DINOv2 feature centroids. We therefore do not claim a geometry winner on global tree fidelity.

The clean claim our experiments support is narrower than “geometry matters globally”: hyperbolic latent structure adds local-retrieval value over the encoder while Euclidean latent structure does not, and that local advantage is robust to which of three reference trees defines “local.”

Related work. Hyperbolic prototype embeddings are an established tool for tree-shaped data (Nickel & Kiela, 2017; Sala et al., 2018; Ganea et al., 2018). Khrulkov et al. (2020) report a similar asymmetry on few-shot benchmarks (helps neighborhood structure, leaves top-1 unchanged); we confirm that pattern on a hand-built taxonomy with a tree-aware regularizer, and quantify it against linear and k -NN calibration baselines that previous reports of the same asymmetry do not include. Tree-distance-matrix penalties over learnable points are used implicitly in hyperbolic hierarchical clustering (Chami et al., 2020); here we use one as an explicit, tunable regularizer.

2. Data and reference trees

We use WikiArt-Refined (Tan et al., 2019), $\sim 81,446$ paintings labelled with one of 27 styles, with the supplied 70/30 train/val split. The corpus is heavily class-imbalanced (133.3 $\times$ ratio between Impressionism with 13,060 exam-

ples and Analytical Cubism with 77; full distribution in Appendix B). Two design choices follow: the prototype classifier has no per-class bias (only geometric distance enters the logit), and we report balanced accuracy alongside top-1.

Reference trees. WikiArt ships no hierarchy and art-history offers several credible taxonomies, so we use three. The *default* tree (Section A) follows standard art-historical lineage. The *chronological* tree groups styles into six era buckets (Renaissance, Baroque–Rococo, 19th century, early 20th century, modern post-war, non-Western). The *flat* tree makes every style a direct child of the root and serves as a deliberate null whose pairwise distances are constant off-diagonal. For a tree T with leaves u, v , tree distance is the unweighted shortest-path edge count $d_T(u, v) = \text{depth}(u) + \text{depth}(v) - 2 \text{depth}(\text{LCA}(u, v))$. We additionally build two *empirical* reference trees by agglomerative clustering of class-mean encoder features (Section 4.2): one from CLIP and one from DINOv2, used only for evaluation.

3. Model and estimation

Likelihood model. Let ϕ be a frozen CLIP ViT-B/16 encoder (Radford et al., 2021) and $g_\theta : \mathbb{R}^{512} \rightarrow \mathcal{M}$ a two-layer MLP with GELU and dropout, where the latent manifold $\mathcal{M}$ is either Euclidean $\mathbb{R}^d$ or a Poincaré ball $\mathbb{B}_c^d$ of curvature $c > 0$. For an image x of style $y \in \{1, \dots, K\}$, set $z = g_\theta(\phi(x))$, and let $\{\mathbf{p}_k\}_{k=1}^K \subset \mathcal{M}$ be learnable class prototypes. The class-conditional likelihood is a softmax over distance to prototype on $\mathcal{M}$:

$$p(y = k \mid z; \{\mathbf{p}_k\}) \propto \exp(-d_{\mathcal{M}}(z, \mathbf{p}_k)).$$

The encoder feeds the head identically in both geometries; switching geometries changes exactly two things: the head’s final transform (identity for $\mathbb{R}^d$; $\exp_0^c(u) = \tanh(\sqrt{c} \|u\|) u / (\sqrt{c} \|u\|)$ for the ball) and the metric used by the classifier:

$$d^c(x, y) = \frac{1}{\sqrt{c}} \operatorname{arccosh}\left(1 + \frac{2c \|x - y\|^2}{(1 - c \|x\|^2)(1 - c \|y\|^2)}\right).$$

Backbone, MLP width, dropout, batch size, optimiser, and schedule are shared.

Tree-structured regularizer. We add a tree-structured regularizer on prototype layouts. Let $D_{ij} = d_{\mathcal{M}}(\mathbf{p}_i, \mathbf{p}_j)$ be the matrix of pairwise prototype distances on $\mathcal{M}$ and $T_{ij} = d_T(i, j)$ the tree distance matrix on the reference tree T . The regularizer penalises deviation between the *shapes* of these two distance matrices,

$$\mathcal{R}(\{\mathbf{p}_k\}) = \frac{\lambda}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left(\frac{D_{ij}}{D} - \frac{T_{ij}}{T} \right)^2,$$

where $\mathcal{P}$ is the set of $\binom{K}{2}$ off-diagonal pairs, $\bar{D}, \bar{T}$ are the means of D, T , and λ controls regularizer strength. Mean-normalising each pair before differencing makes the regularizer invariant to the absolute scale of distances on $\mathcal{M}$: the geometry is free to pick whatever scale the likelihood prefers, while the regularizer constrains only the relative structure of the prototype distance matrix. $\lambda = 0$ recovers the cross-entropy baseline used in prior hyperbolic-image work (Khrulkov et al., 2020). We treat $\mathcal{R}$ as a penalty rather than a Bayesian log-prior: $\mathcal{R}$ is not the log of a normalised density on $\mathcal{M}^K$ (it is mean-rescaled on each evaluation), so it has no proper-prior interpretation, and we report regularized maximum-likelihood estimates rather than MAP estimates of a posterior.

Estimation. We minimise the regularized negative log-likelihood $\mathcal{L}(\theta, \{\mathbf{p}_k\}) = \mathcal{L}_{CE}(\theta, \{\mathbf{p}_k\}) + \mathcal{R}(\{\mathbf{p}_k\})$ by stochastic gradient descent. Adam optimises the MLP head; Riemannian Adam from `geoopt` (Kochurov et al., 2020) optimises the hyperbolic prototypes, projecting each update back onto the manifold. The regularizer is computed once per gradient step over the $\binom{27}{2} = 351$ off-diagonal style pairs, negligible cost next to the per-batch likelihood term. The framing makes explicit that the regularizer is a structured penalty on a geometric quantity (the prototype distance matrix) and that the choice of latent manifold controls how cheaply that penalty can be driven to zero.

Distortion bound. The regularizer is a function of pairwise prototype distances, so in any latent geometry where the tree distance matrix T admits a low-distortion isometric embedding, the regularizer cost at the optimum is small. Sala et al. give distortion bounds $\sim 1/d$ for $\mathbb{B}_c^d$ versus a constant floor for $\mathbb{R}^d$ at fixed d (Sala et al., 2018). The empirical question is whether that asymptotic statement is visible on a real dataset at modest d , against a real classification likelihood, with a finite-sample fit rather than an isometric embedding objective.

Training details. 150 runs total; batch size 4096; Adam $\eta = 10^{-3}$ for the head and 10^{-2} for the prototypes; weight decay 10^{-4} ; dropout 0.1; gradient clipping at norm 1.0; 30 epochs. CLIP features are pre-cached as float16 tensors, keeping each run under 10 seconds on a single Apple M-series GPU and the full sweep under half an hour. Every headline configuration is replicated across three seeds; reported error bars are seed standard deviation. Significance comparisons across seeds use paired- t tests on the 18 (dim, seed) pairs of the dimension sweep, with hyperbolic configurations selected at the best curvature per pair.

Evaluation. A useful style embedding can be useful along three orthogonal axes, and we report metrics for each. *Classification* is top-1, top-5, and balanced accuracy. *Global tree*

fidelity is the Spearman correlation between the learned prototype distance matrix and the tree distance matrix, plus mean and worst-case multiplicative distortion. *Local tree preservation* is sibling and cousin recall@ k among the k nearest neighbours of each validation embedding, for $k \in \{5, 10\}$. Sibling/cousin sets are derived from a reference tree; we evaluate against the default, the CLIP-empirical, and the DINOv2-empirical tree to test sensitivity (Sections 4.2 and 4.5).

4. Results

Three findings emerge from the 150-run sweep, summarised here and developed below. (i) Hyperbolic prototypes recover local tree structure substantially better than matched Euclidean ones, robustly across reference-tree construction (Section 4.2). (ii) Calibrated against linear and k -NN baselines on raw CLIP features, the hyperbolic fit is the only trained model that adds structural value over the encoder; the Euclidean fit matches the encoder on classification but adds nothing on retrieval (Section 4.3). (iii) The global tree-fidelity comparison is metric- and reference-tree-dependent and we do not claim a winner there (Section 4.6).

4.1. Headline comparison

Selecting the best configuration of each geometry by mean top-1 yields Table 1: the two geometries split along a local-vs-global axis. Euclidean leads classification (top-1 +5.3 pp, top-5 +2.4 pp) and global-tree Spearman against the default tree (+0.108); hyperbolic leads sibling recall@5 (+0.052) and cousin recall@5 (+0.047). The seed bands do not overlap on any of the five metrics. The remaining subsections show that this split is structural rather than the artifact of any single configuration.

4.2. Local tree structure

The local-recall difference is the strongest signal in the sweep. On the default tree (Figure 2), hyperbolic sibling recall@5 stays near 0.19 across $d \in \{2, 4, 8, 16, 32, 64\}$, while Euclidean sibling recall decreases from 0.221 at $d=2$ to 0.142 at $d=64$. This monotone divergence with embedding capacity is consistent with the geometric prediction: in $\mathbb{R}^d$ the distortion of the tree distance matrix grows with the available capacity to spread classes for likelihood, while in $\mathbb{B}_c^d$ the curvature absorbs that pressure. Aggregated over 18 paired seed configurations (6 dimensions $\times$ 3 seeds, hyperbolic at the best curvature per pair), the mean gap is +8.7 pp on sibling recall and +15.2 pp on cousin recall, with sign agreement 0.94 on each and paired- t $p < 10^{-4}$.

The sibling/cousin sets used by recall@5 are themselves a function of which reference tree we use, so a sceptical reading would treat the local advantage as an artifact of

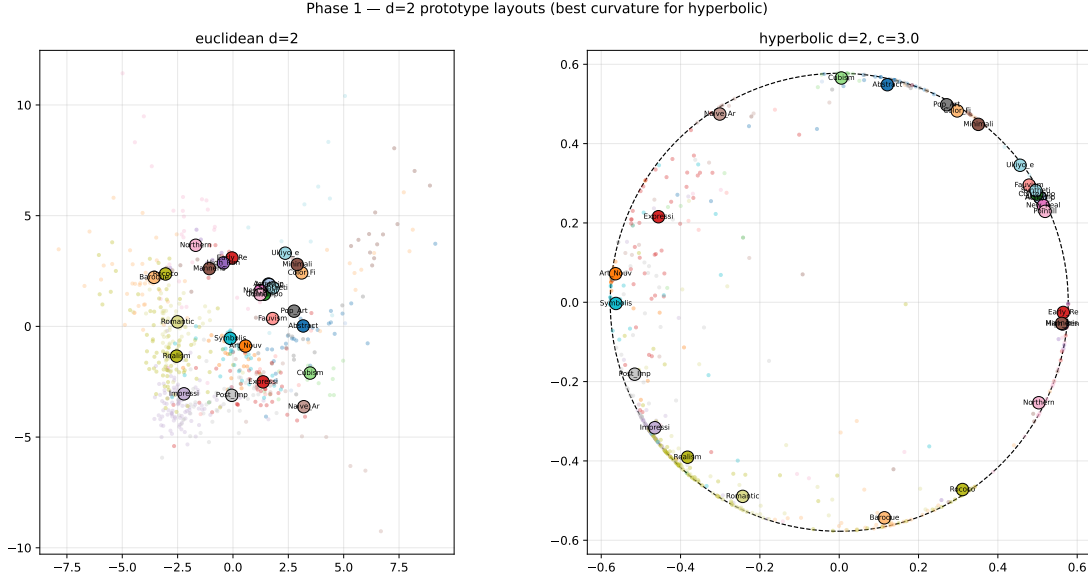


Figure 1. Prototype layouts at $d=2$, best curvature per geometry. Each large dot is a learned style prototype; small dots are a 600-image sample of validation embeddings. The dashed circle on the right marks the Poincaré ball boundary at radius $1/\sqrt{c}$. Hyperbolic prototypes settle into a near-boundary ring (mean radius 0.561, std 0.013 across all 27 styles, against a boundary at 0.577). This is the exponential-volume regime where the manifold has the capacity for trees (Nickel & Kiela, 2017; Sala et al., 2018). Euclidean prototypes scatter diffusely with no analogous structural pressure (mean radius 2.96, std 0.995). The geometry behaves as theory predicts; the rest of the paper measures whether that geometric difference shows up in downstream metrics.

the hand-built tree. We test this by rebuilding sibling and cousin sets from data-driven *empirical* trees: agglomerative average-linkage clustering on class-mean encoder features, converted to integer-edge tree distances. Figure 3 shows the comparison against a CLIP-derived empirical tree; the hyperbolic lead is preserved and slightly larger (+6.0 pp at $d=8$ versus +4.6 pp on the default tree). Replacing the empirical tree’s construction encoder with DINOv2 (Oquab et al., 2024) replicates the lead at every $d \geq 4$ (full numbers in Section F). The local advantage of hyperbolic prototypes is reference-tree independent in the strict sense that it survives three distinct reference-tree constructions, including two encoder-derived ones that share no construction step with the hand-built tree.

4.3. Calibration baselines

A reader cannot judge a 65.9% top-1 number without external calibration. Two reference classifiers train directly on raw frozen CLIP features, with no learned head and no prototype objective: logistic regression and k -NN-5. Logistic regression achieves 64.1% top-1 and 61.1% balanced accuracy; k -NN achieves 63.5% top-1, 58.8% balanced accuracy, and sibling recall@5 of 0.160/0.366 on default-tree / CLIP-empirical-tree relations.

Read against these baselines: the best Euclidean configuration (65.9% top-1) beats logistic regression by 1.8 pp; on balanced accuracy it underperforms (55.4% vs. 61.1%).

Best-top-1 Euclidean’s sibling recall@5 is 0.136, below the k -NN baseline of 0.160. The best-top-1 hyperbolic configuration (60.6% top-1) sits 2.9 pp below k -NN on classification but 2.8 pp above it on sibling recall (0.188 vs. 0.160). At $d=8$ specifically, hyperbolic sibling recall@5 reaches 0.195 on the default tree and 0.416 on the empirical tree, against the k -NN baseline’s 0.160/0.366. Figure 4 plots the full top-1-vs-sibling-recall scatter: the Euclidean cluster sits where k -NN already lives; only hyperbolic configurations populate the upper region. On this dataset and encoder, the hyperbolic fit is the only trained model that adds local-structural value over the encoder.

4.4. Regularizer strength

The relative ordering of the two geometries is preserved across the regularizer-strength sweep $\lambda \in \{0, 0.1, 0.3, 1, 3\}$ at $d \in \{8, 16\}$, $c=1$ (Figure 5). At $\lambda=0.1$ the regularizer Pareto-improves both: Euclidean $d=8$ top-1 rises 0.7 pp to 65.0% while tree-Spearman rises from 0.292 to 0.345; hyperbolic gains a fraction of a point on top-1 and 2 pp on sibling recall. At $\lambda=3$ both geometries reach tree-Spearman ≈ 0.7 but classification collapses (Euclidean 46.9%, hyperbolic 37.4%): the regularizer succeeds at the metric it was constructed to optimise, at substantial likelihood cost. The Euclidean top-1 lead and the hyperbolic sibling-recall lead persist at every λ tested.

metric	euclidean	hyperbolic	logistic-on-CLIP	kNN-5-on-CLIP
Top-1	0.659 $\pm$ 0.003	0.606 $\pm$ 0.002	0.641	0.635
Top-5	0.959 $\pm$ 0.001	0.935 $\pm$ 0.001	—	—
Balanced acc.	0.554 $\pm$ 0.002	0.463 $\pm$ 0.004	0.611	0.588
Class-center / tree Spearman	0.350 $\pm$ 0.027	0.242 $\pm$ 0.002	—	—
Avg. tree distortion	1.742 $\pm$ 0.009	1.861 $\pm$ 0.003	—	—
Worst tree distortion	4.733 $\pm$ 0.253	7.147 $\pm$ 0.076	—	—
Dendrogram F1	0.034 $\pm$ 0.030	0.000 $\pm$ 0.000	—	—
Sibling recall@5	0.136 $\pm$ 0.003	0.188 $\pm$ 0.004	—	0.160
Cousin recall@5	0.249 $\pm$ 0.003	0.296 $\pm$ 0.005	—	0.277

Table 1. Best Euclidean vs. best hyperbolic configuration across the dimension and regularizer-strength sweeps, selected per geometry by mean top-1 across seeds. Mean $\pm$ seed standard deviation. Two rightmost columns: logistic regression and k -NN-5 on raw CLIP features (no learned head, no prototype objective). Euclidean leads classification and global-tree metrics over hyperbolic prototypes; hyperbolic prototypes lead sibling and cousin recall by 4–5 pp. Read against the calibration baselines, the Euclidean fit is tied with logistic on top-1 and with k -NN on sibling recall; only the hyperbolic fit improves on k -NN for sibling and cousin recall.

Scaling with d (hyperbolic = best curvature per d)

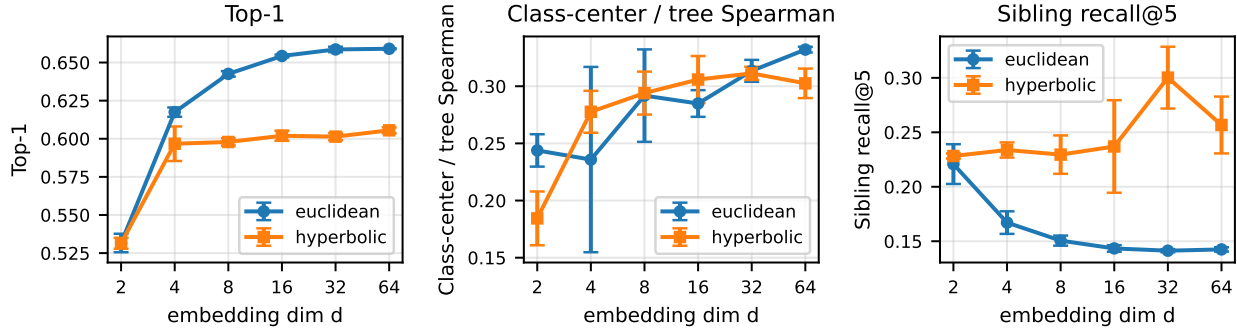


Figure 2. Metrics versus embedding dimension d (90 runs; three seeds per config; hyperbolic at the best curvature per d per seed). Top-1 saturates near $d=16$ for both geometries with Euclidean ahead by 4–6 pp. Sibling recall@5 decreases with d for Euclidean (from 0.221 at $d=2$ to 0.142 at $d=64$) but stays nearly flat for hyperbolic, opening a 5 pp gap by $d=64$.

4.5. Training-tree and class-imbalance robustness

Re-training each geometry against the chronological and flat reference trees (with $d=8$, $c=1$, $\lambda=1$, evaluated against the default tree throughout) does not change the geometry winner on any panel: Euclidean wins top-1 on every training tree, hyperbolic wins sibling recall on every training tree (Figure 6). Inverse-frequency class-weighted training preserves the geometry gap on every metric we report (top-1 remains 5 pp Euclidean, sibling recall remains 4 pp hyperbolic; Section G). Substituting DINOv2 features into the empirical-tree construction reverses the direction of the small global-Spearman gap but preserves the hyperbolic local-recall lead on every reference tree (Section F).

4.6. Global tree fidelity is unstable

The third finding is partly a negative result. Aggregated paired- t tests over the full dimension sweep give $p = 0.68$ for class-center / default-tree Spearman and a sign agreement of exactly 0.50. The small headline gap ($\rho_{\text{Eu}} - \rho_{\text{Hy}} \approx 0.108$ in Table 1) is therefore selecting a noisy outlier

through best-of-sweep. Mean tree distortion does favour Euclidean significantly ($+0.040$, $p < 10^{-4}$), so the two natural global-fidelity metrics disagree about which geometry wins. Replacing the hand-built reference with a CLIP-derived empirical tree reverses the Spearman direction at every $d \geq 4$ (hyperbolic ahead by $+0.01$ to $+0.05$); replacing the empirical tree’s construction encoder with DINOv2 reverses it back (Euclidean ahead by up to $+0.06$). Hand-built and encoder-derived trees correlate only weakly ($\rho \approx 0.08, 0.12$ on pairwise distances); CLIP-empirical and DINOv2-empirical correlate 0.50. Across all four reference-tree constructions we tried, the only stable claim about global tree fidelity is that the comparison is not stable. The local claim survives every variation.

5. Discussion

The clean finding is that, in our tree-regularized prototype model on a real hierarchical-classification problem, the latent manifold has a local effect on tree fidelity even when its global effect is unstable. Hyperbolic prototype geome-

Local recall metrics under default vs empirical tree sibling/cousin sets

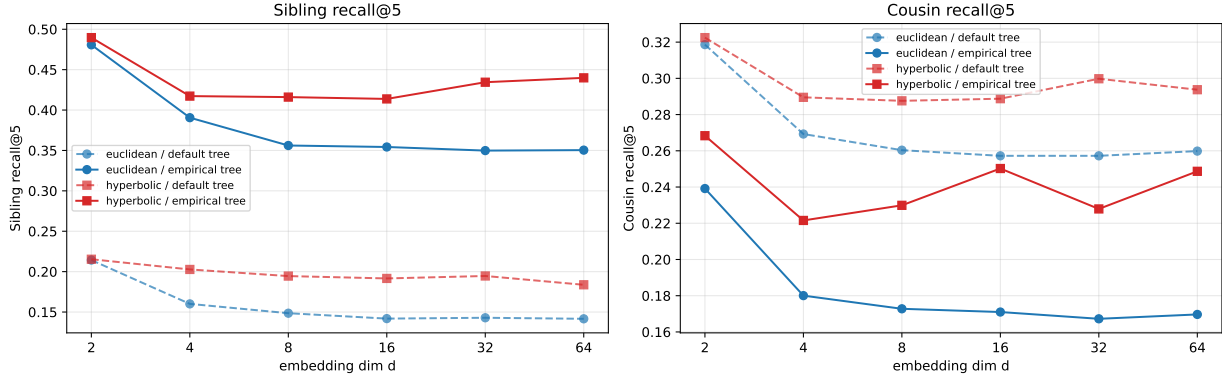


Figure 3. Sibling and cousin recall@5 with relations defined by the default tree (dashed) and a CLIP-derived empirical tree (solid), per geometry per embedding dimension. The hyperbolic lead is preserved under the empirical-tree redefinition; on sibling recall the gap widens slightly. The DINOv2-derived empirical tree is in Section F.

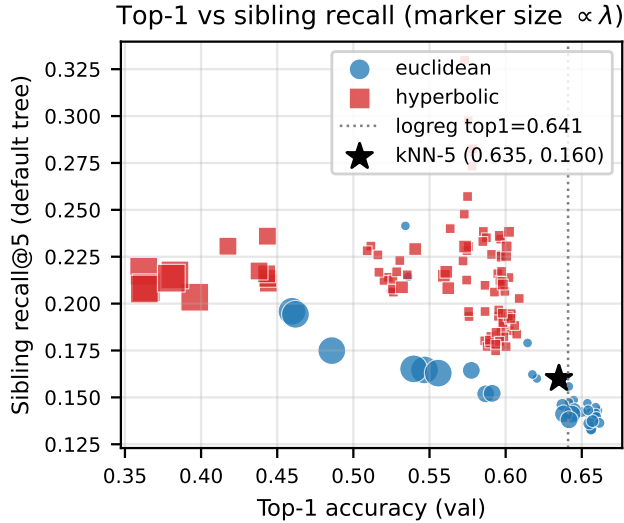


Figure 4. Top-1 accuracy versus sibling recall@5 for every sweep configuration. Marker size scales with the regularizer strength λ . Vertical dotted line: logistic-regression top-1 on raw CLIP features (64.1%). Black star: k -NN-5 on raw CLIP features (sibling recall 0.160). The Euclidean cluster sits at or below the encoder baseline on retrieval; only the hyperbolic cluster populates the upper region of the plot.

try is the only one of our trained models that meaningfully improves on the frozen encoder for local hierarchy preservation. Sibling recall@5 at $d=8$ is 0.195 for hyperbolic, 0.149 for Euclidean, and 0.160 for k -NN-5 on raw CLIP features: the Euclidean fit is tied with the encoder; the hyperbolic fit improves on it by 3.5 pp. The advantage holds when sibling sets are redefined from data-driven empirical trees built from CLIP or DINOv2 features, and aggregates to +8.7 pp on sibling and +15.2 pp on cousin recall over the full dimension sweep with paired- t $p < 10^{-4}$.

The classification gap is real but uninformative about ge-

ometry. A logistic-regression head on raw CLIP features achieves 64.1% top-1, which the Euclidean fit matches at $d=8$ (64.3%) and the hyperbolic fit underperforms by 4–6 pp. Neither fit is doing meaningful classification work that a linear head on the same features cannot; the difference between the two fits on this axis is best read as “Euclidean prototype distances do not distort classification beyond what the encoder already supports, while hyperbolic prototype distances do.”

The global tree-fidelity comparison is the place where careful framing matters most. Mean tree distortion favours Euclidean significantly against the default tree; class-center / tree Spearman is not significantly different from zero across the sweep ($p = 0.68$); the small global-Spearman gap reverses sign between empirical reference trees built from CLIP and from DINOv2 feature centroids. “Either geometry wins on global tree fidelity” is not a claim our experiments support. Sibling and cousin recall favour hyperbolic on every reference tree we tried.

Limitations. We cannot generalise beyond medium-scale, Western-canon-heavy WikiArt with a frozen CLIP-ViT-B/16 encoder for training; the local advantage we identify may not survive a fine-tuned encoder or a hierarchy substantially deeper than the 27-leaf taxonomy used here. We cannot conclude that the hyperbolic geometry does substantive work inside the head MLP, since only the prototypes live on the manifold.

Extensions. Three natural follow-ups in increasing cost. A genuinely hyperbolic head along the lines of Möbius layers (Ganea et al., 2018) would let the manifold shape the representation rather than only the decision boundary, and is the cheapest test of whether the classification gap closes. The empirical reference trees in this paper are built

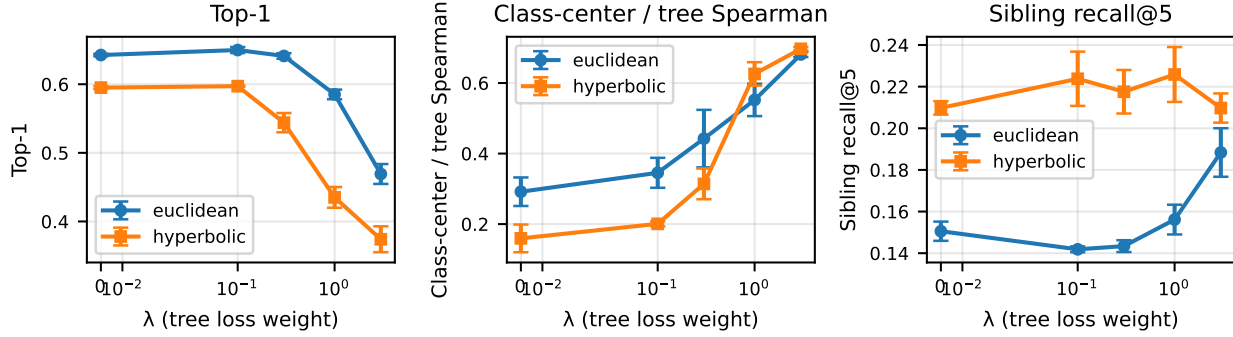
Hierarchy-aware loss at $d=8$, $c=1.0$ 

Figure 5. Top-1, prototype-tree Spearman, and sibling recall@5 versus regularizer strength λ at $d=8$, $c=1$ (three seeds per point). The regularizer drives tree-Spearman from ~ 0.3 to ~ 0.7 in both geometries, but classification collapses for $\lambda \geq 1$. The hyperbolic top-1 gap never closes; sibling recall is stable for hyperbolic across all λ , while for Euclidean it rises only at the largest tested λ , where classification has already collapsed. Mean tree distortion (omitted here for legibility) decreases uniformly with λ in both geometries; full four-panel version in Section D.

Sensitivity of geometry winner to training tree

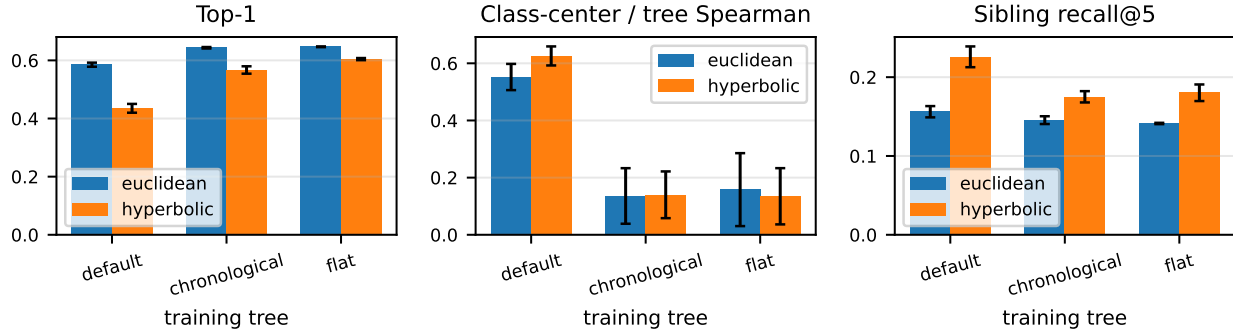


Figure 6. Each geometry trained with $\lambda=1$ at $d=8$ against three different reference trees, then evaluated against the default tree. Bars are seed means; whiskers are seed standard deviation. Tree-Spearman against the default tree collapses for non-default training trees; the regularizer only helps the metric anchored to the training tree. The geometry winner does not change on any panel: Euclidean wins top-1 on every tree, hyperbolic wins sibling recall on every tree. Full four-panel version including mean tree distortion in Section E.

in Euclidean space from encoder features. Building one in hyperbolic space, or by direct tree-learning methods (Chami et al., 2020), would close a remaining circularity. Replication on a deeper hierarchy (iNaturalist with WordNet, or a fine-grained subdivision of WikiArt’s largest classes) would test the limits of the local advantage we observe.

Broader impact

The system itself is small (a 27-way classifier on cached features), but a few of its design choices have implications worth being explicit about. The default reference tree is built from a Western, lineage-based art-history canon. Every metric in this paper that mentions “the hierarchy” is anchored to that taxonomy, which is implicitly endorsed by anyone using the numbers. Our tree-variant experiments are partial mitigation: they document how much each conclusion depends on the choice of tree. A deployment in a museum

or classroom should treat the tree as configuration, not as a constant. WikiArt is heavily biased toward European painting; the corpus has 27 styles but only one (Ukiyo-e) sits outside the European-and-American canon, and East Asian ink-painting traditions that span centuries are collapsed into that single label. A retrieval system trained on this data will under-rank non-Western works for ambiguous queries. Hyperbolic geometry, which tightens local neighborhoods, could compound the effect by making those already-tight neighborhoods more confident. We therefore do not recommend deploying hyperbolic style embeddings for attribution or authentication tasks without expert human review: 50–65% top-1 accuracy over 27 well-known styles is far below the threshold any serious provenance, insurance, or legal decision should require.

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A. Default style hierarchy

The default reference tree, hand-built from standard art-history references and used as the primary evaluation reference throughout the paper.

```

Root
|-- Early_Renaissance
|   |-- Northern_Renaissance
|   |-- High_Renaissance
|       |-- Mannerism_Late_Renaissance
|       |-- Baroque
|       |-- Rococo
|       |-- Romanticism
|           |-- Realism
|               |-- Contemporary_Realism
|               |-- Impressionism
|                   |-- Post_Impressionism
|                       |-- Pointillism
|                       |-- Fauvism
|                       |-- Cubism
|                           |-- Analytical_Cubism
|                           |-- Synthetic_Cubism
|-- Symbolism
|   |-- Art_Nouveau
|   |-- Expressionism
|       |-- Abstract_Expressionism
|           |-- Action_painting
|           |-- Color_Field_Painting
|           |-- Minimalism
|-- Pop_Art
|   |-- New_Realism
|-- Ukiyo_e
|-- Naive_Art_Primitivism

```

B. Class distribution and tree-distance matrices

C. Curvature within the hyperbolic family

The body Figure 2 shows the dimension scaling for both geometries; here we add detail on curvature. Sweep configuration: $\text{geometry} \times d \in \{2, 4, 8, 16, 32, 64\} \times c \in \{0.1, 0.3, 1, 3\}$ for hyperbolic, three seeds. Within the hyperbolic family, curvature interpolates between the two extremes: higher c pulls prototypes closer to the boundary of the disk, where the Poincaré metric becomes most curved; this exchanges 2–3 pp of top-1 for 4–5 pp of sibling recall. At $d=8$, the optimal c for top-1 is 0.3 (59.8%) and the optimal c for sibling recall is 3.0 (0.230). This is consistent with how hyperbolic capacity is distributed: most volume lies near the boundary, where the leaves of a tree should live.

D. Regularizer-strength sweep, full panel set

E. Training-tree ablation, full panel set

The body Figure 6 summarises the three-tree ablation; the underlying observation is that training against a tree other than the evaluation tree provides essentially no useful hierarchical signal: tree-Spearman against the default tree drops to ≈ 0.14 for both non-default training trees, and top-1 climbs back to the $\lambda=0$ baseline because the regularizer no longer competes with the likelihood.

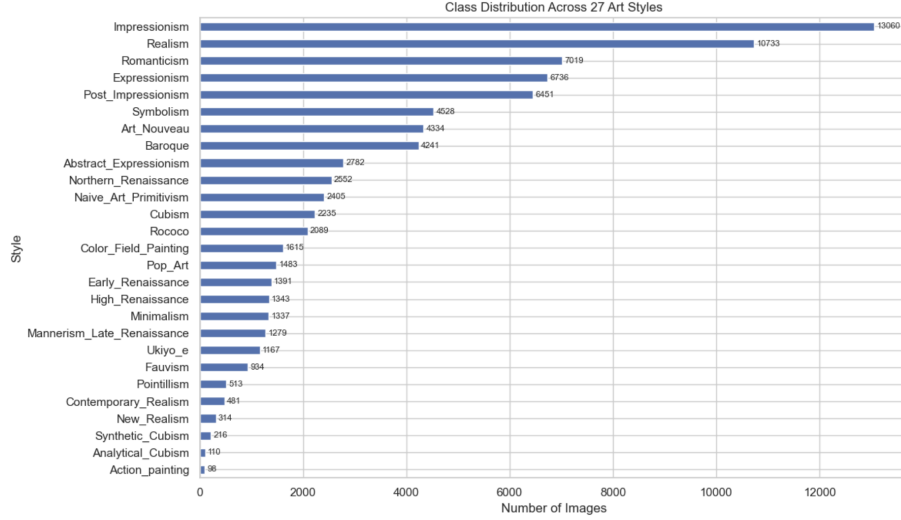


Figure 7. Per-style image counts on WikiArt-Refined. Impressionism dominates with 13,060 examples; Analytical Cubism, Action Painting, and Synthetic Cubism each have fewer than 250. The imbalance ratio between extrema is $133.3\times$.

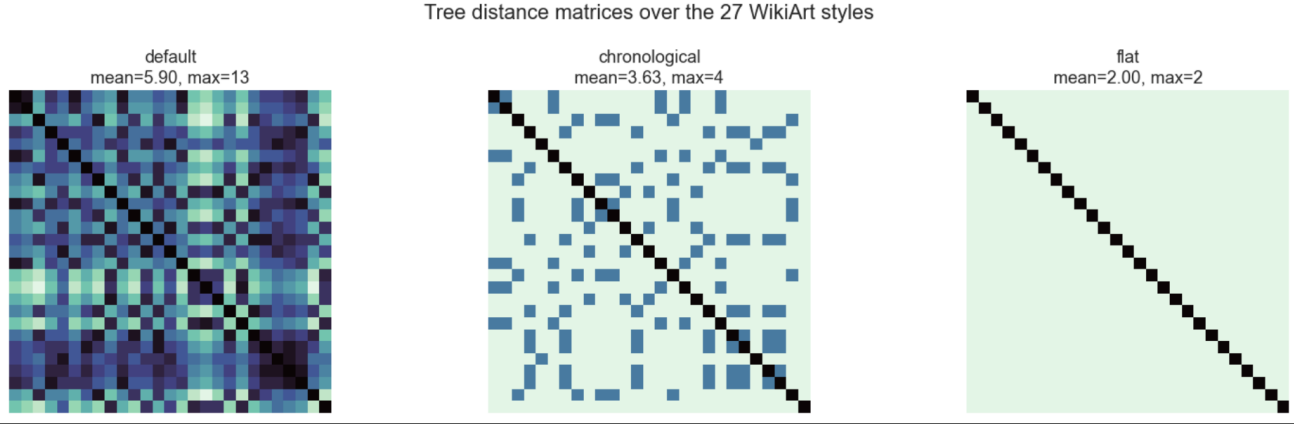


Figure 8. Tree distance matrices for the three reference hierarchies used in this paper: default lineage (left), chronological era grouping (centre), flat null (right). Rows and columns are styles in the same order across panels.

F. Empirical reference trees and DINOv2 cross-encoder

Tree-vs-tree. Pairwise-distance Spearman between hand-built default and CLIP-empirical is 0.08; between default and DINOv2-empirical it is 0.12; between CLIP-empirical and DINOv2-empirical it is 0.50. Whatever the hand-built lineage tree captures, the encoders see only a weak projection of it; whatever the encoders capture, they capture it similarly. The hand-built tree is the outlier.

Empirical sibling and cousin recall. Sibling/cousin sets used by recall@5 in the body figures are derived from STYLE_HIERARCHY (the default tree). To test reference-tree sensitivity we rebuild sibling and cousin sets from each empirical tree’s binary linkage (siblings: the leaves in the other branch of the leaf’s first merge; cousins: the leaves in the uncle subtree at the grandparent merge) and rerun the recall computation. The hyperbolic lead is preserved on every reference tree (sibling recall@5 at $d=8$, default / CLIP-empirical / DINOv2-empirical: Eu 0.149/0.356/0.322, Hy 0.195/0.416/0.380).

Global-Spearman direction is encoder-specific. Against the DINOv2-empirical tree, Euclidean prototypes lead at every $d \geq 4$, with $\rho_{\text{Eu}} - \rho_{\text{Hy}}$ growing to +0.06 at $d=64$ (0.413 vs. 0.357). This is a negative result for the strongest reading of the CLIP-empirical finding and motivates the global-fidelity caveat in the body.

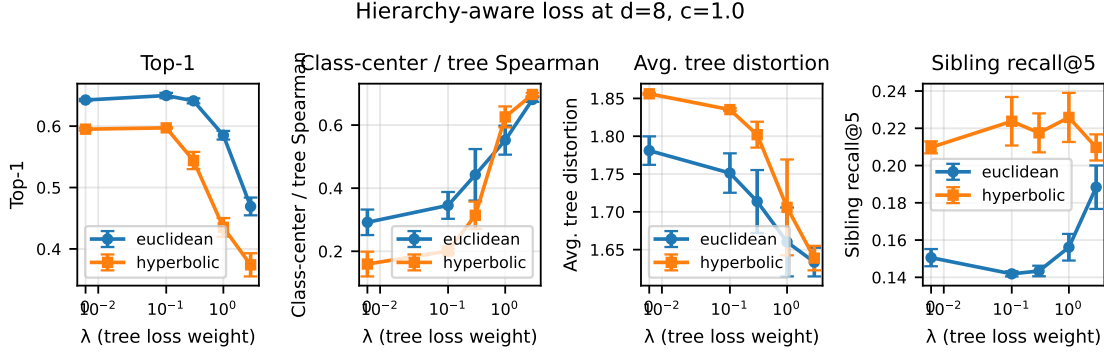


Figure 9. Full four-panel version of the regularizer-strength sweep (body Figure 5 drops the third panel for legibility). Mean tree distortion decreases monotonically with λ in both geometries, mirroring the rise in tree-Spearman.

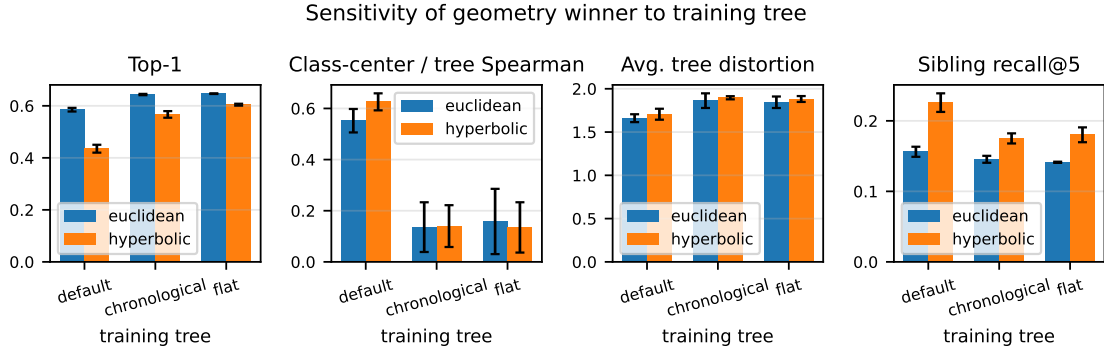


Figure 10. Full four-panel version of the training-tree ablation (body Figure 6 drops the third panel for legibility). Tree distortion against the default tree is uniformly higher when training against a non-default tree.

G. Class-imbalance robustness

WikiArt is heavily skewed (Impressionism has $133\times$ more training examples than Action Painting). The default sweep uses unweighted cross-entropy; rerunning the $d=8$ winners of each geometry with inverse-frequency class-weighted cross-entropy (three seeds) shifts absolute numbers as expected (top-1 drops by roughly 6 pp for both: Eu $64.3 \rightarrow 58.3\%$, Hy $59.8 \rightarrow 53.4\%$; while balanced accuracy rises sharply: Eu $56.4 \rightarrow 65.6\%$, Hy $44.5 \rightarrow 62.1\%$), but the geometry gap is preserved on every axis. Top-1 remains 5 pp Euclidean, sibling recall remains 4 pp hyperbolic, and class-center / tree Spearman shifts by less than a hundredth. The balanced-accuracy gap narrows from 11.9 to 3.5 pp, suggesting hyperbolic’s main classification weakness in the default setup was disproportionately on rare classes; it remains in Euclidean’s favour.

H. Confusion structure

I. Reproducibility

The full sweep CSV records, for every config, the training hyperparameters and every evaluation metric (150 rows). To reproduce:

```
python scripts/sweep.py --phase 1 --device mps # ~14 min, 90 configs
python scripts/sweep.py --phase 2 --device mps # ~10 min, 48 configs
python scripts/sweep.py --phase 3 --device mps # ~3 min, 18 configs
python scripts/baselines.py # logreg, kNN baselines
python scripts/significance.py # paired-t, sign tests
python scripts/make_figures.py # regenerate figures
```

Code, sweep configurations, and exact metric implementations will be released after de-anonymisation.

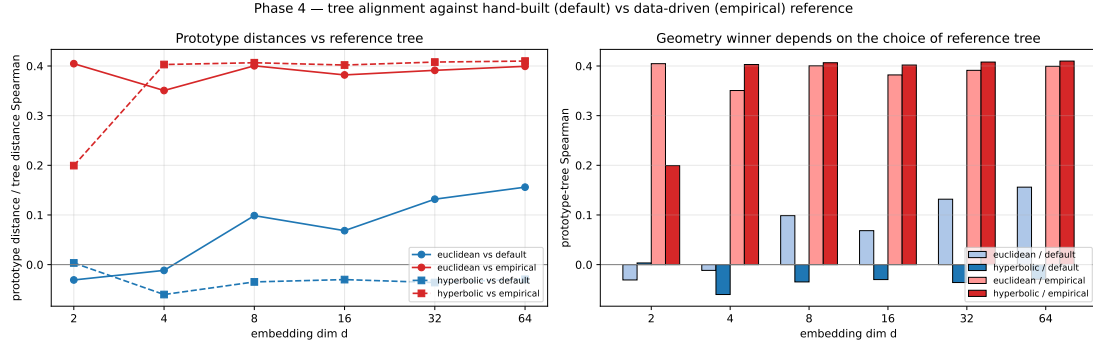


Figure 11. Prototype-distance / tree-distance Spearman against the hand-built default tree (blue) and a CLIP-empirical tree (red). Both geometries’ prototypes align substantially better with the empirical tree than with the hand-built one. Caveat: the empirical tree is built in Euclidean space from the same CLIP features used to train the prototypes, so the alignment with prototype distances is partly expected by construction. The geometry comparison is the load-bearing claim, not the absolute level.

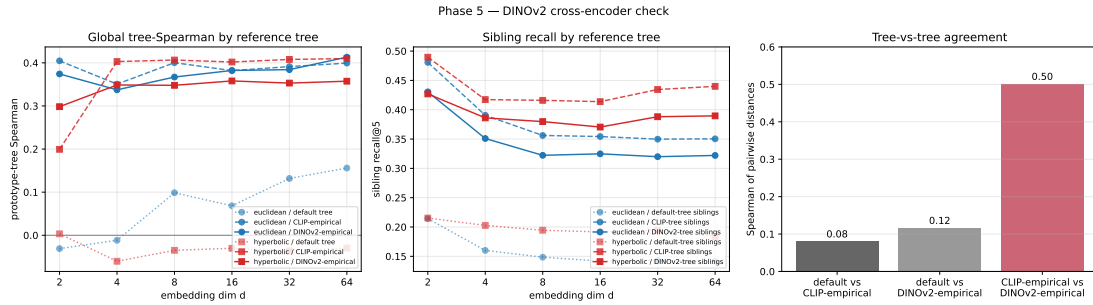


Figure 12. Substituting a DINOv2 ViT-B/14 encoder for CLIP in the empirical-tree construction. **Left:** prototype-tree Spearman against three references. The hyperbolic global-Spearman lead seen against the CLIP-empirical tree does not replicate against the DINOv2-empirical tree: Euclidean prototypes lead at every $d \geq 4$. **Centre:** sibling recall@5 against the three reference-tree sibling-set definitions. Hyperbolic leads on every reference at every $d \geq 4$. **Right:** pairwise-distance Spearman between reference trees. Hand-built correlates only weakly with either encoder-derived tree (0.08, 0.12); the two encoder-derived trees correlate 0.50 with each other.

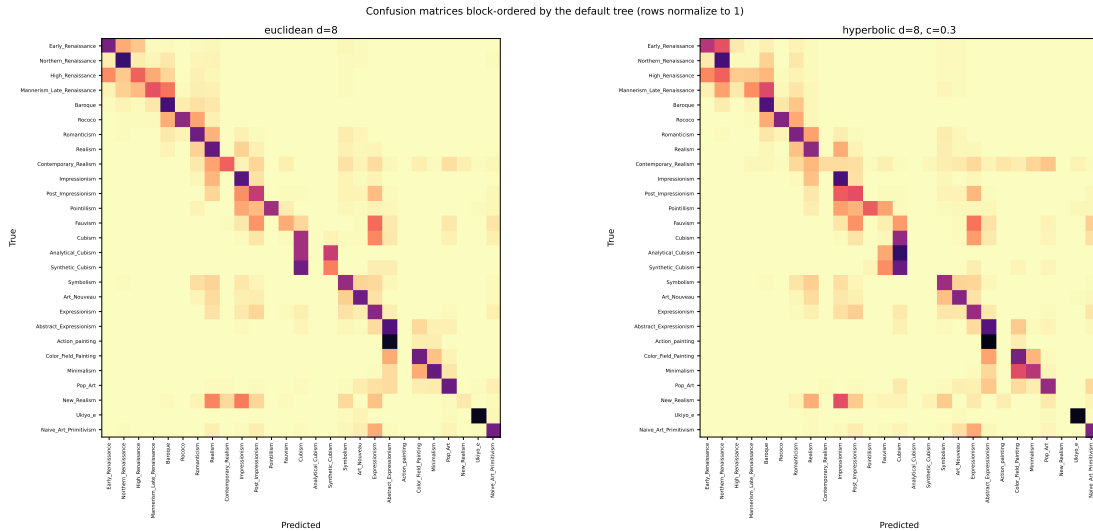


Figure 13. Row-normalized confusion matrices at $d=8$, with rows and columns permuted by a depth-first traversal of the default style tree so that hierarchically adjacent styles sit next to each other on both axes. Both models concentrate mistakes in near-diagonal blocks: errors predominantly fall on tree-adjacent styles. The hyperbolic block structure is visibly softer near the diagonal: more confusion mass spreads into immediate siblings, less mass jumps to distant styles. This is the same fact the sibling-recall numbers report, in a form a viewer can absorb at a glance.